

Paper A-13: Earth Plate Tectonics: A Constraint-Driven Flux Dynamics (CDFD) Perspective

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Abstract

This paper asks whether Constraint-Driven Flux Dynamics (CDFD) and Adaptive Flux Limitation (AFL) can give the named system a sharper audit language. The working variables are driving flux Φ , constraint C , surface responsiveness S , and structural memory M_s . The useful question is plain: what can be measured, what would count as overload, and what result would make the mapping fail?

1 Series Context

Part I sets the flow–constraint language used here [1]. Part II develops the Life Number and the origins-of-life use of the same notation [2]. Part III applies AFL to biology and medicine [3]. Part IV is the cross-domain stress test: it asks where the notation clarifies measurement, and where it becomes only a metaphor.

2 Introduction

Notation. In Part IV, C names the active constraint or capacity burden in the system being discussed. Φ is the driving flow, S is the surface response, and M_s is retained structural memory. Λ appears only where the Part II Life Number is being referenced [2]. Other Greek symbols are local coefficients whose meaning is set by the equation in which they appear.

2.1 The Mujjabi Universal Capacity Law

Building directly upon the Fundamental Physics (Part I) [1], the **Mujjabi Universal Capacity Law** proposes that across all scales—from black hole event horizons to global financial markets and power grids—driving high flux (Φ) against a rigid topological constraint (C) can drive system saturation. When Φ/C approaches critical thresholds, the system does not scale linearly; it undergoes a topological phase transition.

2.2 The Mujjabi Universal Memory Law

The **Mujjabi Universal Memory Law** frames persistent systemic collapse. When a complex network fails to clear its structural memory (M_s) and its relaxation parameter decays ($\tau_{relax} \rightarrow \infty$), temporary stressors become permanent rigidities. This is the proposed CDFD mechanism behind ecological death, economic depressions, and infrastructural gridlock.

The Earth's crust is not static; it is a continuously evolving mosaic of rigid plates floating atop a viscous, convecting mantle. Where these plates meet, immense stress accumulates along fault zones. Earthquakes occur when the accumulated stress exceeds the frictional strength of the fault, resulting in a sudden, violent displacement.

In CDFD, tectonic fault lines are not merely mechanical interfaces; they are dynamic regulatory constraints across a massive thermodynamic network. Like a clogged biological artery or a bottlenecked supply chain, a locked fault is a system struggling to process its requisite throughput.

3 The Tectonic Adaptive Surface

We govern the fault mechanics using the Universal Adaptive Ratio:

$$\Psi_s = \left(\frac{\Phi}{C} \right) \cdot S \cdot M_s \quad (1)$$

In the context of plate boundaries:

- **Flux (Φ):** The continuous kinetic driving force generated by mantle convection and slab pull pushing the tectonic plates together or past one another.
- **Constraint (C):** The frictional resistance of the fault line, specifically the interlocking rock asperities that prevent smooth sliding.
- **Responsiveness (S):** The elastic capacity of the surrounding crust to deform and absorb stress without breaking.
- **Memory (M_s):** Fault locking. As the fault remains stationary over centuries, chemical bonding and mechanical interlocking fuse the rock faces together. The fault "remembers" its static position, driving $M_s \rightarrow 1.0$ and cementing the constraint.

3.1 The Seismic Cycle as a Regime Shift

During the interseismic period, the tectonic flux (Φ) continues unceasingly from the mantle. However, because the fault is locked by its memory (M_s) and friction (C), the physical flow of the crust halts at the boundary. The system enters a state of extreme localized constraint ($\Psi_s \ll 1$).

To avoid total stagnation, the system's responsiveness (S) flexes: the crust elastically bends, storing the incoming flux Φ as potential energy. Over decades, this elastic capacity S approaches its ultimate physical limit (the yield strength of the rock).

When S saturates, the adaptive ratio crashes. The system can no longer sustain the incoming flux. The stored topological tension causes a sudden, catastrophic failure of the memory term M_s . The interlocking asperities snap. The constraint C instantly drops to near zero (dynamic sliding friction), and the stored potential energy violently converts back into kinetic flux Φ —an earthquake.

4 Measurement Layer

The first job in any domain is to choose proxies before drawing conclusions. Φ may be a rate of material, energy, information, capital, or signal flow. C is the bottleneck that slows or redirects that flow. S measures the system’s ability to reconfigure under stress, and M_s records the past load that still changes present behavior. A useful application gives those four terms observable definitions, a time scale, and a failure condition. If a standard domain model explains the same data more cleanly, the CDFD/AFL mapping should give way.

4.1 Current Runtime Diagnostic: Universal Network Cascade

The diagnostic script `Part_E_Synthesis/supplementary/run_partiv_discovery.py` keeps this paper tied to the current CDFD Runtime [4], including RK4 field updates, surface dynamics, structural-memory diffusion, and a finite-value audit. In the May 2026 run, the localized hub-drive stress test ended with hub $\Psi_s = 14.8$, peak network $\Psi_s = 14.8$, 21 nodes above $\Psi_s > 1.5$, and 37 maximum memory-locked nodes. I treat those numbers as a runtime sanity check and as a target for later domain measurement, not as a substitute for field data.

5 Conclusion

Earthquakes are mathematically indistinguishable from stellar core collapse or metabolic phase shifts. They represent the abrupt topological resetting of an over-constrained system forced to shed its chronic memory to restore a sustainable flow regime. By applying CDFD, seismology transitions from a descriptive science of elastic rebound into a universal thermodynamic physics.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Release-local runtime diagnostics are available under each Part’s `outputs/` folder, with release-wide synthesis artifacts under `Part_E_Synthesis/supplementary/`, `Part_E_Synthesis/outputs/`, and `Part_E_Synthesis/figures/`.

6 Reference Layer

References

- [1] Steve Bico Mujjabi. Cdfd part i: Fundamental physics - constraint-driven field theory and flux dynamics, 2026.
- [2] Steve Bico Mujjabi. Cdfd part ii: Origins of life and tri-regime bioenergetics - constraint-driven flux dynamics, 2026.
- [3] Steve Bico Mujjabi. Cdfd part iii: Adaptive flux limitation biology and medicine - constraint-driven flux dynamics, 2026.
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